

REPORT

Trader Performance vs Bitcoin Market Sentiment

Assignment for Web3 Trading Team (Hyperliquid)

1. Introduction

In this project, I studied how Bitcoin market sentiment affects trader performance on the Hyperliquid platform.

Market sentiment is measured using the **Fear and Greed Index**, which shows whether the market is feeling:

- Extreme Fear
- Fear
- Neutral
- Greed
- Extreme Greed

The main goal of this assignment is to understand:

- Do traders perform better during fear or greed?
- Does trader behaviour change depend on market mood?
- What insights can help create smarter trading strategies?

2. Datasets Used

This assignment uses two datasets:

2.1 Bitcoin Sentiment Dataset

This dataset provides daily sentiment information with:

- Date
- Classification (Fear/Greed level)
- Sentiment Score (0–100)
- Timestamp

2.2 Hyperliquid Trader Dataset

This dataset contains detailed trading history such as:

- Trader account

- Coin traded
- Trade size (USD)
- Buy/Sell side
- Closed Profit or Loss (Closed PnL)
- Trading fees
- Leverage

3. Data Preparation

To combine both datasets, I followed these steps:

- Converted timestamps into proper date format
- Extracted trade dates from the trader dataset
- Merged both datasets using the date column
- Only 6 trades had missing sentiment values, so they were removed since they were very few compared to the total dataset

After merging, each trade included a sentiment label (Fear/Greed).

4. Analysis Performed

I grouped the data by sentiment category and calculated:

- Average profit per trade
- Total profit contribution
- Win rate (percentage of profitable trades)
- Average trade size
- Average fee paid

This helps compare trader performance under different market moods.

5. Main Results

The summary below shows performance across sentiment regimes:

classification	Avg_PnL	Total_PnL	Win_Rate	Avg_Trade_Size	Avg_Fee
Extreme Fear	198.46	128401.15	31.53	6539.69	1.08
Extreme Greed	33.59	21499.10	16.56	3061.83	0.41
Fear	79.54	519809.90	38.82	34584.93	5.78
Greed	58.09	222293.74	34.72	32256.56	6.56
Neutral	136.08	316259.49	49.87	27182.41	5.11

6. Key Insights

Insight 1: Extreme Fear Has the Highest Average Profit per Trade

Even though Extreme Fear has a low win rate (31.5%), it shows the highest average PnL per trade.

This suggests:

- Fewer trades are profitable
- But the profitable ones tend to be very large winners

This pattern indicates that strong price reversals during panic conditions can lead to high returns for traders who time entries well.

Insight 2: Neutral Market Conditions Are the Most Consistent

Neutral sentiment has:

- The **highest win rate (almost 50%)**
- A strong average PnL
- Moderate trade size and fees

This suggests that traders perform **most consistently** when the market is calm and less emotional.

Insight 3: Fear and Greed Lead to Larger Trade Sizes

Both **Fear** and **Greed** regimes show very large average trade sizes (above \$30,000).

This indicates:

- Traders take larger positions when emotions are strong
- Fear and Greed both encourage aggressive trading behavior

However, this does not always translate to better performance.

Insight 4: Extreme Greed Performs the Worst

Extreme Greed shows:

- The **lowest average PnL**
- The **lowest win rate**
- Very small trade sizes

This suggests that during extreme optimism:

- Market may already be overextended
- Late entries result in poor performance

Insight 5: Higher Fees During Emotional Markets

Fees are highest during **Fear and Greed** regimes.

This happens because:

- Larger trade sizes
- Higher trading frequency

High fees can significantly reduce net profitability.

7. Trading Recommendations

Based on these insights, the following strategies are recommended:

Recommendation 1: Be Selective During Extreme Fear

Extreme Fear offers high reward opportunities, but with low consistency.

- Suitable for experienced traders
- Requires strict risk management

Recommendation 2: Favor Neutral Markets for Stable Performance

Neutral periods provide:

- High win rates
- Balanced risk and reward

This is ideal for systematic and lower-risk strategies.

Recommendation 3: Control Position Size During Fear and Greed

Since Fear and Greed lead to very large trade sizes:

- Apply position size limits
- Monitor fees closely

Recommendation 4: Avoid Overconfidence During Extreme Greed

Extreme Greed shows the weakest performance.

- Reduce exposure
 - Avoid chasing trends late
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6. Conclusion

This analysis shows that market sentiment strongly affects both trader performance and behaviour.

Key conclusions:

- High profits can occur during Extreme Fear, but with low success rates
- Neutral markets offer the most consistent performance
- Emotional market conditions increase trade size and fees
- Extreme Greed is the least favourable environment for trading

Overall, using sentiment-aware risk controls can help traders improve decision-making and profitability.
